

# Yue Ma

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## PROFILE

Postdoctoral researcher building LLM agent systems and evaluation infrastructure for science and engineering, with a focus on decision trajectories, action spaces, and resource-bounded experimentation.

## EDUCATION & ACADEMIC POSITIONS

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| <b>Postdoctoral Researcher</b> , Halicioğlu Data Science Institute, UC San Diego | Mar 2026 — Present  |
| <b>Ph.D. in Particle Physics</b> , University of California, San Diego           | Oct 2020 — Feb 2026 |
| <b>B.S. in Physics</b> , Nanjing University                                      | Aug 2016 — Jun 2020 |

## AGENTIC SYSTEMS, EVALUATION, AND RESEARCH INFRASTRUCTURE

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| <b>Resource-Aware Multi-Agent System for Autonomous Research</b> | Feb 2026 — Present |
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**SIDERIUS**. A closed-loop agentic system for proposing, implementing, and evaluating machine-learning experiments under online compute constraints. Sole architect and developer; initially demonstrated on scientific time-series denoising and now extending to additional domains.

- **Composable Agent Runtime**: Designed six reusable agent capabilities with typed, caller-independent contracts, separating capability implementation from workflow wiring and orchestration policy.
- **Scientific Evaluation and Resource Control**: Built declarative task and evaluation contracts, device-specific runtime and GPU-memory calibration, quality gates, bounded execution, and resumable long-horizon experimentation under explicit compute budgets; demonstrated end to end on the [TIDMAD benchmark](#).

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| <b>SciTra: Scientific-Agent Evaluation and Trajectory Infrastructure</b> | Jul 2026 — Present |
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Founder and lead of a collaboration with approximately 100 scientists across 20+ institutions, developing scientific-agent evaluations and decision-trajectory infrastructure.

- **Scientific Agent Evaluation**: Developing a general methodology for representing decision trajectories, action spaces, outcomes, and resource use, so that agent runs can be compared across agents, tasks, and domains.
- **Trajectory Infrastructure**: Built cross-vendor infrastructure capturing expert—agent interaction, tool use, Git changes, failures, costs, outputs, and typed decision records, with append-only declarations, frozen evidence, and deterministic export for reproducible evaluation.

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| <b>FrontierPhysics: Benchmark for AI Agents in Frontier Physics Research</b> | Jul 2026 — Present |
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Core team member on a benchmark, built with the [BenchFlow](#) team, evaluating how AI agents carry out frontier physics research iteratively, from literature review through research-plan implementation, on tasks that take PhD-level researchers weeks.

- **Evaluation Strategy**: Co-developing evaluation for open-ended, PhD-level physics research tasks, including task and rubric design, success criteria, anti-hacking checks, and evaluation of intermediate research decisions rather than final answers alone.
- **Task Infrastructure**: Contributing infrastructure for reproducible task packaging, agent execution, and validation across physics domains.

## MACHINE LEARNING RESEARCH

### Adaptive Neural Processes for Extreme Forecasting

Nov 2024 — Jun 2025

Probabilistic forecasting under distribution shifts and limited observations. [arXiv:2602.04609](https://arxiv.org/abs/2602.04609)

- **Model Development and Evaluation:** Conceived and derived AdaCNP, a target-conditioned context-reweighting mechanism enabling few-shot adaptation to unseen extreme regimes without retraining; achieved a 22% reduction in MSE and the best negative log-likelihood among Neural Process baselines across two real-world power-system datasets.

### Multi-Fidelity Surrogate Modeling for Scientific Simulation

Jul 2024 — Oct 2024

Learning efficient surrogates for computationally expensive, high-variance detector simulations.

- **Compute-Efficient Scientific ML:** Developed a multi-fidelity surrogate combining Gaussian Processes and Neural Processes to emulate high-dimensional rare-event simulations, reducing simulation cost by 96.7% while improving background-suppression efficiency by 66.5%; contributed core algorithms and derivations to the [ICLR 2025 Spotlight paper “RESuM”](#) (acknowledged for significant contribution).

## STATISTICAL INFERENCE AND LARGE-SCALE SCIENTIFIC COMPUTING

Research within the [XENONnT Collaboration](#), a 200+ scientist international experiment operating petabyte-scale data systems for dark-matter and solar-neutrino searches.

### Probabilistic Modeling and Statistical Inference (Corresponding Author)

Oct 2020 — Present

One of five corresponding authors on XENONnT's first measurement of solar pp neutrinos at ultra-low momentum transfer.

- **Inference and Experimental Evaluation:** Developed Bayesian and maximum-likelihood inference frameworks, improving the precision of efficiency estimates by 60% through new calibration and data-selection strategies; built Monte Carlo workflows for uncertainty propagation, sensitivity estimation, and hypothesis testing.

### Data Infrastructure and High-Throughput Computing

Jan 2024 — Present

- **Distributed Data Systems:** Key maintainer of the open-source [strax](#) and [straxen](#) frameworks processing petabyte-scale detector data for 200+ collaborators; reduced ETL latency by 40% through vectorized NumPy and Pandas workflows, Numba acceleration, and memory-efficient processing.

## TECHNICAL SKILLS

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|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming &amp; ML</b>         | Python, PyTorch, NumPy, Pandas, SciPy, Scikit-learn, C/C++, Bash, Linux, Git                                                                                 |
| <b>Agents &amp; Evaluation</b>      | LLM agent systems, multi-agent orchestration, agent evaluation, benchmark design, action-space and trajectory instrumentation, autonomous experimental loops |
| <b>Systems &amp; Infrastructure</b> | Pydantic, FastAPI, pytest, GitHub Actions, SLURM/HPC, ETL pipelines, resource monitoring and performance optimization                                        |

## SELECTED PUBLICATIONS

- Aprile, E. *et al.* (2026). “First Measurement of Solar Neutrinos through Elastic Neutrino-Electron Scattering at the keV Scale”. [arXiv:2608.29450](https://arxiv.org/abs/2608.29450). **One of five corresponding authors.**
- Chenxi Hu, **Yue Ma**, Yifan Wu, Yunhe Hou (2026). “Resilient Load Forecasting under Climate Change: Adaptive Conditional Neural Processes for Few-Shot Extreme Load Forecasting”. [arXiv:2602.04609](https://arxiv.org/abs/2602.04609).
- Aprile, E. *et al.* (2023). “First Dark Matter Search with Nuclear Recoils from XENONnT”. *Phys. Rev. Lett.* **131**, 041003.